MILITARY SPECIFICATION SHEET

ELECTRON TUBES, RECEIVING

TYPES OB3 AND OB3A 1/
a b

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

DESCRIPTION: Voltage regulator

Outline	a	---	12-7 (EIA)	Ⓑ
	b	---	9-7 (EIA)	Ⓑ
Base	a	---	B6-3	
	b	---	B6-8	
Envelope	a	---	ST12	
	b	---	T9	
Cathode	a, b	---	Glow discharge	

Base connections:

Pin No.	a, b	---	1	2	3	5	7	8
Element		---	nc	k	7 (see note 1)	a	3 (see note 1)	nc

ABSOLUTE-MAXIMUM RATINGS:

Parameter:		Ebb	Starting voltage	Operating voltage	Operating current	Alt
Unit:		Vdc	Vdc	Vdc	mAdc	ft
Maximum:	a, b	---	---	90 (approx)	30	10, 000
Minimum:	a, b	---	130	---	5	---
TEST CONDITIONS:	a, b	150	---	---	--- (see note 2)	---

GENERAL:

Qualification - OB3 - Not required
OB3A - Required

1/ To identify those tests that are applicable to a given type or to several types; tube types are designated by letters.

Ⓑ denotes changes

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METHOD	REQUIREMENT OR TEST	TYPE	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS		UNIT
							MIN	MAX	
	Quality conformance inspection, part 1								
3347	Ionization voltage (1)	a, b	Rp/Ib = 5 to 30 mAdc (see notes 3 and 4)	0.65	II	Ez	---	125	Vdc
3337	Voltage drop (1)	a, b	Rp/Ib = 5 mAdc	0.65	II	Etd(1)	80	---	Vdc
3337	Voltage drop (2)	a, b	Rp/Ib = 30 mAdc	0.65	II	Etd(2)	---	100	Vdc
3335	Regulation	a, b	Etd(2) - Etd(1)	0.65	II	Reg	---	6.0	Vdc
3278	Noise and oscillation (oscillation test)	a, b	Esig = 100 mVac; RL = 500; Ebb/Ib = 5 to 30 mAdc	0.65	II	---	---	---	---
3278	Noise and oscillation (noise)	a, b	Ebb/Ib = 30 mAdc; RL = 500	0.65	II	ΔEB	---	15.0	mVac
1201	Short and discontinuity detection	a, b		0.4	II	---	---	---	---
	Quality conformance inspection, part 2								
1031	Low-frequency vibration	a, b	No voltage	6.5	S3	---	---	---	---
3305	Leakage current	a, b	Eb = 50 Vdc	6.5	S3	LIb	---	10.0	μAdc
1101	Secureness of base and base insert	a, b		6.5	S3	---	---	---	---
1111	Base pin solder depth	a, b	(See note 5)	6.5	S3	---	---	---	---
1105	Permanence of marking	a, b		---	---	---	---	---	---
	Quality conformance inspection, part 3								
---	Life-test provisions	a, b	Group A; Rp/Ib = 30 mAdc	---	---	---	---	---	---
---	Life-test end points (500 hours)	a, b	Ionization voltage	---	---	Ez	---	130	Vdc
			Voltage drop (1)	---	---	Etd(1)	77	---	Vdc
			Voltage drop (2)	---	---	Etd(2)	---	103	Vdc
			Regulation	---	---	Reg	---	9.0	Vdc

NOTES:

1. Pins 3 and 7 tied internally.
2. Fixed resistor may be used and Ebb varied to give desired current.
3. Test to be conducted with the tube exposed to light intensity of 5 to 50 foot candles.
4. This test to be performed at the conclusion of the holding period.
5. This test applies to flexible leads as well as for rigid leads.

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Custodians:
Army - EL
Navy - EC
Air Force - 11

Review activities:
Army - EL
Navy - SH
Air Force - 11, 85
DSA - ES

User activities:
Army - ME
Navy - AS, OS, MC, CG
Air Force - 19

Preparing activity:
Navy - EC

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